03-DerivRich-exc

Construct differentiation formulae

1. Pelle is unhappy that at the first point a of $[a,b]$ he has to use simple forward differences to approximate the derivative $f'(a) = D_h f(a) + \mathcal{O}(h)$. Instead, he extrapolates by Richardson to weigh D_h and D_{2h} and obtains an $\mathcal{O}(h^2)$–formula for the derivative $f'(a)$: f63751

$$\hat{D}_h f(a) = (\alpha D_h + \beta D_{2h}) f(a) = f'(a) + \gamma h^2 + \mathcal{O}(h^3).$$

   (a) Which α and β does Pelle use? Find also the exact coefficient γ for the leading h^2–error term.

   (b) Use Pelle's operator $\hat{D}_h$ at step $h = 0,01$ to approximate $f'(1)$ for $f(x) = \ln x$.

2. Let $D_h = \frac{1}{h}(T_h - I)$ and $\bar{D}_h = \frac{D_h + D_{-h}}{2}$ be the forward and central difference operators. 74878b

   (a) Prove that $= D_{-h} D_h = D^2_{\frac{h}{2}}$. Compute also $D^2_{\frac{h}{2}} f(\frac{\pi}{3})$ for $f(x) = \cos(x)$ and $h = 0.01$.

   (b) Estimate the error of approximation for the second derivative $f''(x)$ of a smooth function f,

$$D_{\frac{h}{2}} f(x) - f''(x) = b h^\alpha + \mathcal{O}(h^\beta), \quad \beta > \alpha,$$

   with exact constants b, α and β. Compare with your error (of course, $f''(\frac{\pi}{3}) = -\cos\frac{\pi}{3} = -\frac{1}{2}$) in (a).

3. Let f be a smooth function. One can construct a $\mathcal{O}(h^2)$–approximation formula for its derivative f' by carefully interpolating between two $\mathcal{O}(h)$ formulae, e.g., $f'(x) \approx \tilde{D}_h f(x) \stackrel{\text{def}}{=} \frac{2D_{h/2} + D_{-h}}{3} f(x)$. 9a80e1

   (a) Show that $\tilde{D}_h$ acting on $f(x)$ is an $\mathcal{O}(h^2)$–approximation of $f'(x)$, such that

$$\tilde{D}_h f(x) = f'(x) + \beta h^2 + \mathcal{O}(h^3).$$

   Find the exact constant β expressed through the derivatives of f at x. (Hint: $D_h = \frac{e^{hD} - I}{h}\cdot$)

   (b) Compute $\tilde{D}_h f(0)$ if $f(x) = e^x$ and $h = 0.1$ and verify that the error $\tilde{D}_h f(x) - f'(x) \approx \beta h^2$.

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4. Given the values $f_k = f(x_k)_0^n$ of a smooth function f at evenly spaced nodes $x_k = x_0 + kh$, $k = 0, 1, 2, \ldots, n$, forward differences give the not-so-good $\mathcal{O}(h)$ approximation of the first derivative

$$D_h f(x_0) = f'(x_0) + \mathcal{O}(h), \quad D_h f(x_0) = \frac{1}{h}(T_h - I) f_0 = \frac{f_1 - f_0}{h}.$$

   One obtains a better $\mathcal{O}(h^2)$ approximation of $f'(x_0)$ using the finite difference operator $\tilde{D}_h$:

$$\tilde{D} f(x_0) = \frac{-3I + 4T_h + T_h^2}{2h} f(x_0) = \frac{-3f_0 + 4f_1 - f_2}{2h} = f'(x_0) + \beta h^2 + \mathcal{O}(h^3)$$

   (a) For $f(x) = e^x$ and $h = 0.01$, compute the approximations $D_h f(0)$ and $\tilde{D}_h f(0)$ of the derivative $f'(0) = 1$. State also the errors of approximation you obtained.

   (b) Find also the exact value of the constant β. Compare whether $\tilde{D}_h f(0) - f'(x_0) \approx \beta h^2$ in (a).

5. The forward/backward difference operators $D_{\pm h}$ give an $\mathcal{O}(h)$ approximation of the derivative f' of a smooth function f at a point x, while their average, the central differences operator $\bar{D}_h = \frac{D_h + D_{-h}}{2} = \frac{T_h - T_{-h}}{2h}$, provides an $\mathcal{O}(h^2)$ approximation of f', namely bb5f09

$$|D_{\pm h} f(x) - f'(x)| \simeq \frac{f''(x)}{2!} h = \mathcal{O}(h), \quad \text{whereas} \quad |\bar{D}_h f(x) - f'(x)| \simeq \frac{f'''(x)}{3!} h^2 = \mathcal{O}(h^2).$$

   Kalle weighs up D_h and D_{-h} a bit differently and comes up with the approximation formula

$$f'(x) \approx \tilde{D}_h f(x) \stackrel{\text{def}}{=} (0.4 D_h + 0.6 D_{-h}) f(x).$$

(a) Find the constants a, b and c for which

$$\tilde{D}_h f(x) = \frac{1}{h}(aT_h + bI + c\,T_{-h})f(x) = \frac{af(x+h) + bf(x) + cf(x-h)}{h}.$$

Use afterwards Kalle's formula to approximate $f'(2)$ with $h = 0.1$ for $f(x) = \ln(x)$.

(b) Use that $T_h = e^{hD}$ to find the order α of the error estimate $|\tilde{D}f(x) - f(x)| \simeq Ch^\alpha$ as a function of h.

6. Consider the approximation of the third derivative f''' of a smooth function f at an internal point a: 1b750a

$$f'''(a) = D_h \bar{D}^2_{\frac{h}{2}} f(a) + bh^\alpha + \mathcal{O}(h^{\alpha+1}), \quad \text{where} \quad D_h f(a) = \frac{T_h - I}{h} f(a), \quad \bar{D}_{\frac{h}{2}} = \frac{1}{2}(D_{\frac{h}{2}} + D_{-\frac{h}{2}}).$$

(a) Write down the approximation explicitly: $D_h \bar{D}^2_{\frac{h}{2}} f(a) = c_2 f(a+2h) + c_1 f(a+h) + c_0 f(a) + c_{-1} f(a-h)$.

(b) Find the order α and the exact constant b for the error term bh^α. Hint: $T_h = e^{hD} = \sum_{k=0}^{\infty} \frac{(hD)^k}{k!}$.

7. Let $f(x) = \arctan(x)$ and $x_0 = 1$. Tabulate with 6 significant digits (x_k, f_k), where $f_k = f(x_k) = f(x+kh)$, $h = 0.1$ and $k = -2, -1, 0, 1, 2$. Compute the three approximations for the second derivative $f''(x_0)$: 932a07

(a) $D^2_h f(x_0)$ (b) $D^2_{-h} f(x_0)$ (c) $\bar{D}^2_{\frac{h}{2}} f(x_0)$.

Compare with the actual value of $f''(x_0)$. Which approximation is the most accurate and why?

8. Suppose that f is smooth ($f \in C^5$ or better) in the vicinity of a fix x. Determine the error of the central differences approximation of $f'(x)$ with *exact* orders k and l and constant b_k: 8fc8a3

(a) by using the Taylor formula

$$\epsilon_h(f) = \bar{D}_h f(x) - f'(x) = \frac{f(x+h) - f(x-h)}{2h} - f'(x) = b_k h^k + \mathcal{O}(h^l),\ l \geq k+1.$$

(b) by using the linearity of D, D_h and $\bar{D}_h \stackrel{\text{def}}{=} (D_h + D_{-h})/2$ and work on the operator side of:

$$\epsilon_h(f) = \left(\frac{D_h + D_{-h}}{2} - D\right) f(x), \quad D_h = \frac{e^{hD} - I}{h}. \qquad \text{Hint: } e^t = 1 + t + \frac{t^2}{2!} + \frac{t^3}{3!} + \cdots.$$

(c) Compute by $\bar{D}_h f(0)$ for $f(x) = e^x - x$ och $h = 0.1$ and compare the error with $b_k h^k$ in (a).

9. Let f be a smooth function. Use that the action $D_h f(x) = \frac{e^{hD} - 1}{h} f(x)$ to find an expression for the error fd76b2

$$D^2_h f(x) - f''(x) = \frac{f(x+2h) - 2f(x+h) + f(x)}{h^2} - f''(x) = ah + bh^2 + \mathcal{O}(h^3)$$

with exact constants a and b.

10. Numme Knatte cooks up the formula $D_{-h} D^2_h f(x)$ to approximate $f'''(x_1)$ using the four values $f_k := f(x_k) = f(x_0 + kh)$, $k = 0, 1, 2, 3$. 16d3ee

(a) Spell out the formula $D_{-h} D^2_h f_1$ he should use to compute the approximation of $f'''(x_1)$.

(b) Derive an error estimate $D_{-h} D^2_h f = Ch + \mathcal{O}(h^2)$ with exact constant C.

(c) Use Knatte's formula to approximate $f'''(0.01)$ for $f(x) = \ln(2 + x)$ by using its values f_k at nodes $\{x_k\}_0^3 = \{0.01k\}_{k=0}^3$. Use the values of f_k with at least 10 decimals (Why?).

11. The approximation $\bar{D}_h f(x)$ of $f'(x)$ with central differences of a smooth function $f(x)$ is given by ac6a26

$$\bar{D}_h f(x) = \frac{T_h - T_{-h}}{2h} f(x) = \frac{f(x+h) - f(x-h)}{2h} = f'(x) + b_2 h^2 + \mathcal{O}(h^4).$$

Use Richardson extrapolation to eliminate the $\mathcal{O}(h^2)$-term designing a higher-order formula

$$\lfloor \bar{D}_h : \bar{D}_{2h} \rfloor f(x) = (\alpha \bar{D}_h + \beta \bar{D}_{2h}) f(x) = \frac{1}{h} \sum_{k=-2}^{2} c_k f(x + kh),$$

which doubles up the convergence rate to: $\lfloor \bar{D}_h : \bar{D}_{2h} \rfloor f(x) - f'(x) = \mathcal{O}(h^4)$. Check your formula by approximating $f'(0.7) \approx \lfloor \bar{D}_h : \bar{D}_{2h} \rfloor f(x)$ for $f(x) = \sin(x)$ (x in radians) using $h = 0.01$.

Richardson

1. Knut approximates the derivative of $f(x) = \arcsin x$ at $x = 0.6$ using central differences 6b12c3

$$\bar{D}_h f(0.6) = \frac{f(0.6+h) - f(0.6-h)}{2h} = f'(0.6) + \mathcal{O}(h^\alpha). \quad \text{(Exact value: } f'(0.6) = \frac{1}{\sqrt{1-0.6^2}} = 1.25.\text{)}$$

Initial step is $h_0 = 0.1$, then halved thrice, $h_i = h_0 q^i$, $q = 0.5$, $i = 1, 2, 3$. Still unhappy with the accuracy of $(1.37)10^{-4}$, he performs a 3-fold Richardson extrapolation (below $F = \bar{D}_h f(0.6)$):

$h_0 q^i$	$(q^\alpha)^i$	$\bar{D}_h f(0.6)$	$\mathcal{R}\bar{D}_h f(0.6)$	$\mathcal{R}^2\bar{D}_h f(0.6)$	$\mathcal{R}^3\bar{D}_h f(0.6)$
0.100000	1.000000	1.258994	1.2499381164	1.250000144242	1.2499999999038873
0.050000	0.250000	1.252202	1.2499962675	1.250000002159	
0.025000	0.062500	1.250548	1.2499997687		
0.012500	0.015625	1.250137			

Final error: $\epsilon_3 = |(D - \mathcal{R}^3\bar{D}_h)f(0.6)| = (9.61)10^{-11}$. (Big smile, he squeezed all juice out of his effort!)

(a) A j-fold Richardson applied on $\bar{D}_h f$ differs from the derivative f' of a smooth function f at x by:

$$\mathcal{R}^j\bar{D}_h f(x) - f(x) = Ch^{\alpha_j} + \text{(higher-order terms)} \;\Leftrightarrow\; \epsilon_j = |(D - \mathcal{R}^j\bar{D}_h)f(x)| = \mathcal{O}(h^{\alpha_j}).$$

What orders of convergence α_j , $j = 0, 1, 2, 3$, should Richardson extrapolations exhibit?

(b) Using Knut's computations, find an estimate for α_2, the order of approximation for $\mathcal{R}^2\bar{D}_h f$.
Hint: Error drops ϵ_2: $(1.44)10^{-7} \searrow (2.16)10^{-9}$ with step refinement of h: $0.1 \searrow 0.05$.

Octave/Python

1. Devise a NumPy for routine for computing weighted divided differences $\lfloor \mathbf{d} : \mathbf{f} \rfloor$, taking $\{d_i, f_i\}_{i=0}^n$ and $0 < j < n$ as input parameters and outputs the j:th column of the table $\{\lfloor f_i : f_{i+j} \rfloor\}_{i=0}^{n-j}$ as a np.array: 1455a9

```
import numpy as np
def wDD(di,yi,col=100):                    # Vectorized version
    n=len(di); lastcol = min(n,col+1)      # Choose column
    X,Y = np.array(di),np.array(yi)        # Copy lists as np.arrays
    for j in range(1,lastcol):
        Y = Y[:n-j] + np.diff(Y) * X[:n-j] / ( X[:n-j] - X[j:n] )
    return Y
```

Test by, say, Neville-evaluating the IP $p(x) = x^3$ at $x = 5$ (last line reads $5^3 = 125$ as it should?):

```
n=4; xi=np.arange(n); yi=xi**3;
for j in range(n):
    print(f"# >> Column {j}: {wDD(5-xi,yi,j)}")
```

2. Let $f(x) = \arctan(x)$, $h = 0.2$ and $q = 0.5$. Compute approximations for $f'(1)$ along with the error made (what's the exact value?) at stepsizes $h_i = \{hq^i\}_i^3 = 0$ using: 7b6674

(a) forward differences, $f'(1) = D_h f(1) + b_1 h + b_2 h^2 + \cdots$ and then Richardson extrapolate, $\mathcal{R}^3 D_h f(1)$;

(b) central differences, $f'(1) = \bar{D}_h f(1) + c_2 h^2 + c_4 h^4 + \cdots$ and then Richardson extrapolate, $\mathcal{R}^3\bar{D}_h f(1)$.